Shopping Mall Customer Segmentation Analysis

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1. Introduction

1.1 Project Background

Modern retail environments generate large volumes of customer data. However, without structured analysis, organizations struggle to convert this data into actionable business insights.

This project focuses on analysing shopping mall customer data to understand purchasing behaviour and identify meaningful customer segments. The objective is to enable data-driven marketing strategies that improve customer engagement, retention, and revenue performance.

1.2 Project Objectives

The primary objectives of this project are:

- Clean and prepare customer data for analysis
- Understand demographic and spending behaviour
- Segment customers based on income and spending patterns
- Develop an interactive business dashboard
- Provide strategic business recommendations

1.3 Business Problem

Shopping malls often apply uniform marketing strategies to all customers. This results in:

- inefficient marketing spends
- poor customer engagement
- missed revenue opportunities

Customer segmentation enables targeted decision-making by identifying groups with similar behavioural characteristics.

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2. Dataset Overview

2.1 Dataset Description

The dataset contains customer-level information including:

Variable	Description
Customer ID	Unique customer identifier
Age	Customer age
Gender	Male/Female classification
Annual Income	Customer yearly income
Spending Score	Purchasing behaviour indicator

Total Records: **15,079 customers**

2.2 Data Cleaning & Preparation

The following preprocessing steps were performed:

- Verified dataset structure
- Checked missing values
- Removed duplicate entries
- Standardized column naming
- Converted data types appropriately
- Created derived analytical features

No missing or duplicate records were identified, confirming strong data quality.

2.3 Feature Engineering

Two additional analytical variables were created:

Age Groups

Customers were grouped into business-friendly age ranges:

- 18–25
- 26–35
- 36–45

- 46–55
- 56–65
- 66–75
- 76–90

Boundary validation ensured that minimum age values were properly included.

Customer Segments

Customers were classified into four strategic segments:

- High Income – High Spending
- High Income – Low Spending
- Low Income – High Spending
- Low Income – Low Spending

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3. Exploratory Data Analysis

3.1 Customer Demographics

Analysis revealed:

- Average customer age ≈ **54 years**
- Balanced gender representation
- Wide income distribution across customers

This indicates a mature and stable customer base.

3.2 Spending Behaviour Analysis

Key observations:

- Average spending score ≈ **51**
- Spending behaviour varies independently from income level
- Both high and low income customers demonstrate diverse purchasing habits

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3.3 Gender-Based Insights

Spending behaviour across genders was nearly equal, suggesting:

- marketing campaigns should remain gender-neutral
- behavioural segmentation is more valuable than demographic targeting

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3.4 Age-Based Analysis

Spending behaviour remained relatively consistent across age groups, indicating:

- lifestyle and engagement factors influence purchases more than age alone.

3.5 Income vs Spending Relationship

Correlation analysis showed **minimal relationship** between income and spending score.

Business Interpretation:
Higher income does not automatically translate into higher spending behaviour.

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4. Customer Segmentation Analysis

Customer segmentation formed the core analytical component of this project.

4.1 Segmentation Logic

Customers were classified using mean income and mean spending score thresholds.

This produced four balanced customer clusters.

4.2 Segment Distribution

Each segment contained a comparable number of customers, ensuring meaningful comparative analysis.

4.3 Segment Characteristics

High Income – High Spending

- Premium customers
- Strong purchasing behaviour
- Primary revenue contributors

High Income – Low Spending

- High purchasing capacity
- Under-engaged customer group
- Major growth opportunity

Low Income – High Spending

- Value-driven customers
- Responsive to promotions

Low Income – Low Spending

- Low engagement level
- Long-term development segment

4.4 Data Validation Checks

Professional validation steps were performed:

- Record count verification
- Age group completeness validation
- Segment assignment verification
- Missing value confirmation
- Logical range checks

These checks ensured analytical reliability before dashboard creation.

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5. Interactive Dashboard Development

An interactive Power BI dashboard was developed to communicate insights visually.

5.1 Dashboard Objectives

The dashboard enables stakeholders to:

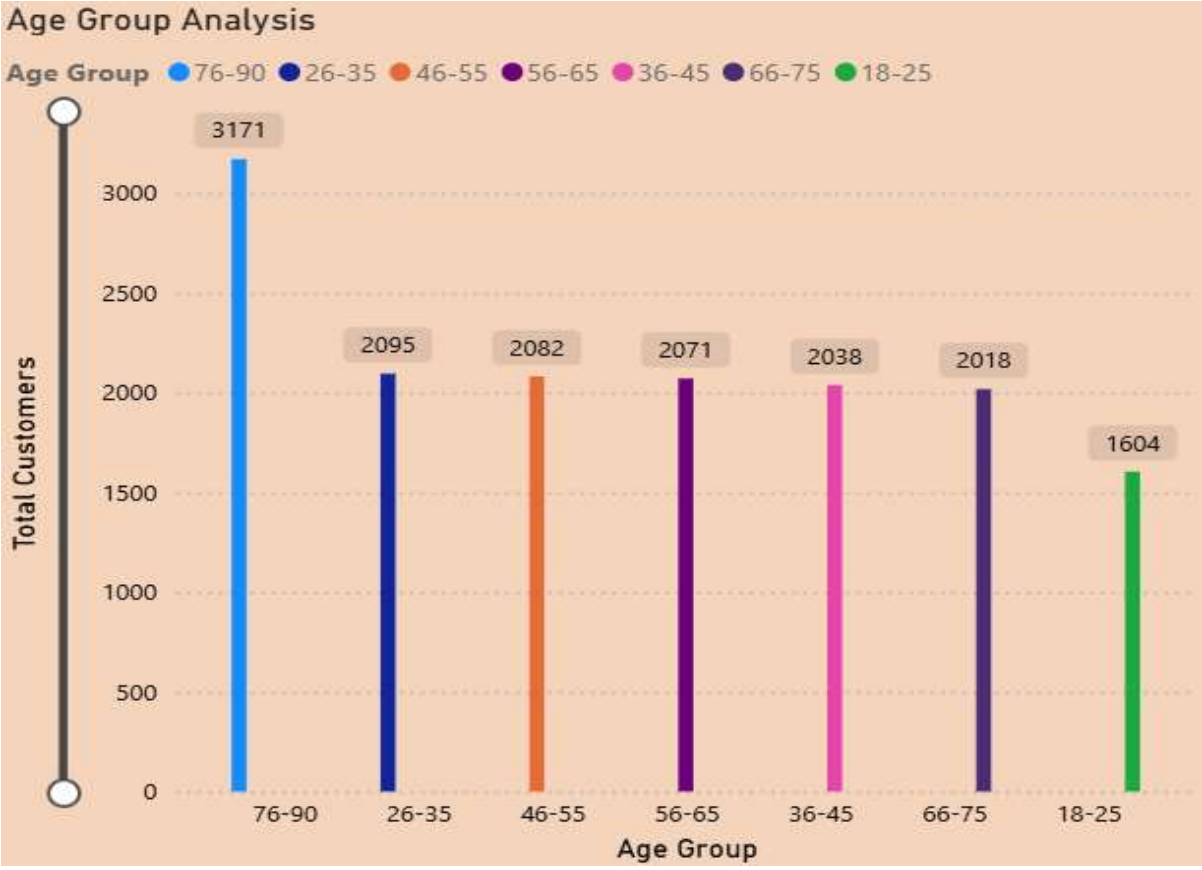
- explore customer demographics
- understand spending patterns
- evaluate customer segments
- support data-driven decisions

5.2 Dashboard Components

Gender-wise Customer Distribution

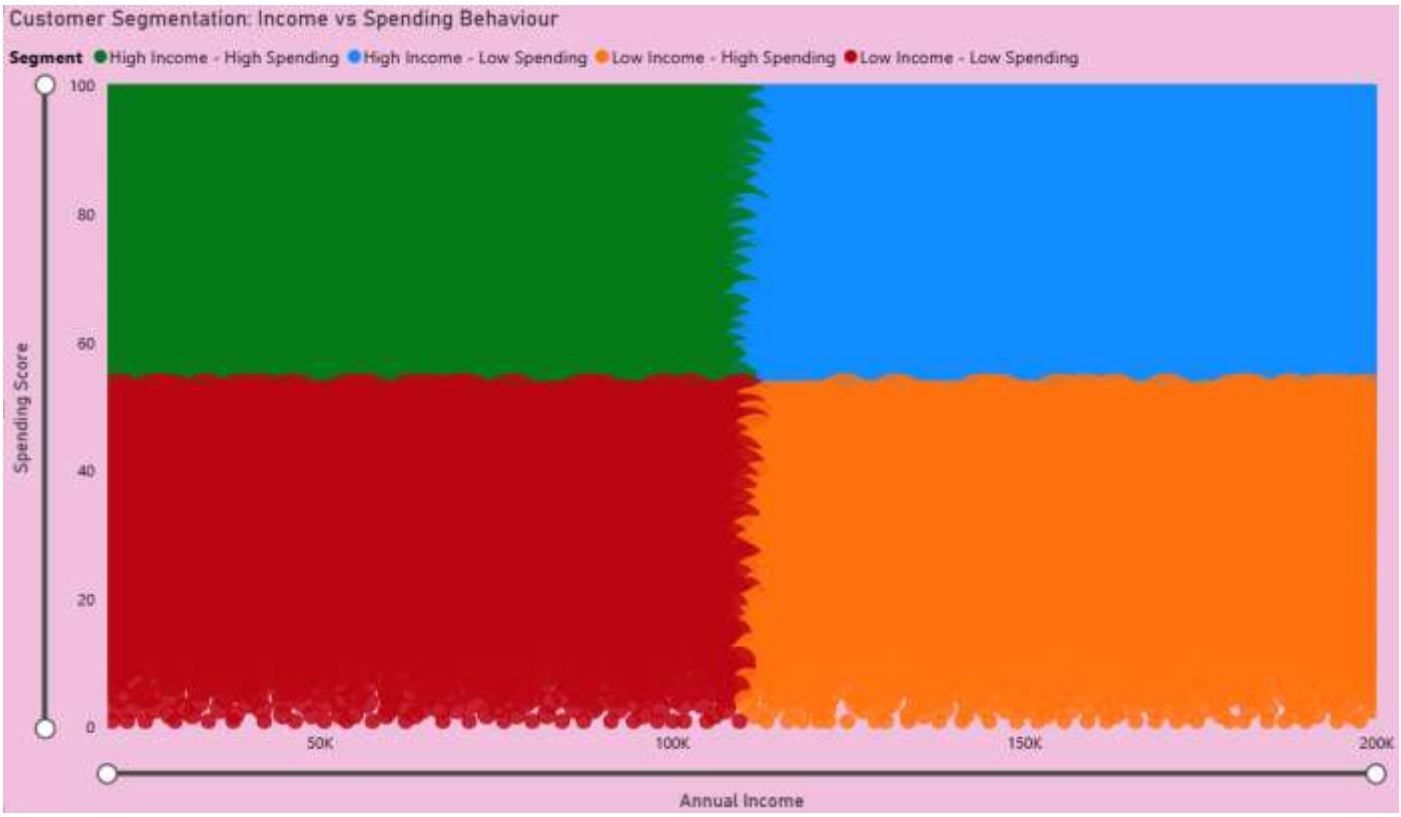
Displays gender composition of the customer base.

Age-wise Customer Analysis



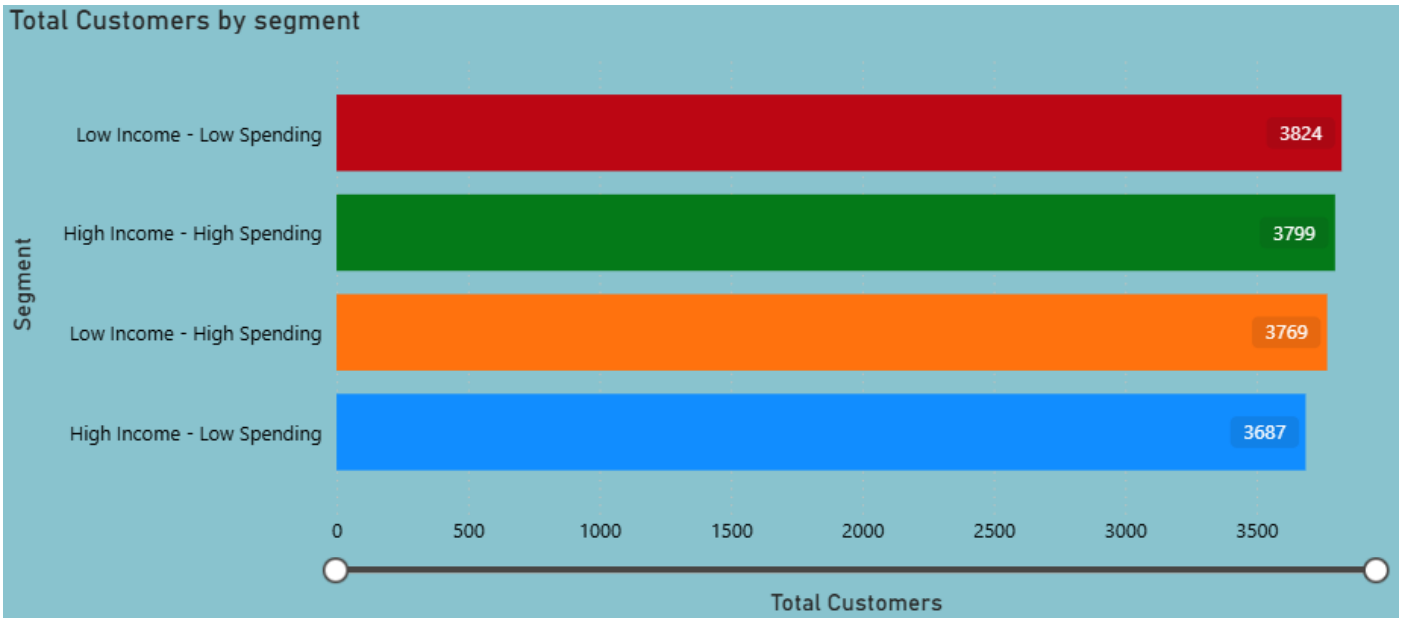
Highlights dominant customer age groups.

Income vs Spending Scatter Plot



Primary analytical visualization showing behavioural clusters and validating segmentation.

Segment-wise Distribution



Shows relative size of each customer segment.

5.3 Dashboard Interactivity

Interactive slicers allow filtering by:

- Gender
- Age Group
- Customer Segment

This enables real-time business exploration.

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6. Business Insights & Recommendations

6.1 Target Segment Priority

The organization should prioritize **High Income – High Spending** customers.

Recommended Actions

- VIP loyalty programs
- Exclusive experiences
- Personalized marketing

Expected Outcome
Higher retention and stable revenue growth.

6.2 Activating Low-Spending Customers

Low-spending customers represent conversion potential.

Recommended Strategies

- Discounts and promotional campaigns
- Bundle pricing
- Reward programs
- Seasonal marketing events

Expected Outcome
Increased engagement and transaction frequency.

6.3 Retaining High-Income Customers

Retention strategies should emphasize experience rather than pricing.

Recommended Strategies

- Personalized recommendations
- Early product access
- Exclusive events
- Premium services

Expected Outcome
Improved customer lifetime value.

6.4 Strategic Business Implications

Key organizational learnings include:

- Behavioural segmentation is more powerful than demographic segmentation.
- Customer engagement drives spending more than income level.
- Data analytics enables efficient allocation of marketing resources.

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7. Business Value Delivered

This project demonstrates how structured analytics transforms raw customer data into strategic business intelligence.

Delivered outcomes include:

- Data-driven customer segmentation
- Interactive decision-support dashboard
- Actionable marketing recommendations
- Improved strategic planning capability

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8. Conclusion

Through systematic data cleaning, exploratory analysis, customer segmentation, and dashboard visualization, this project provides a comprehensive understanding of customer behaviour within a retail environment.

Adopting segmentation-driven strategies enables the shopping mall to transition toward a customer-centric operating model, improving profitability while enhancing customer satisfaction.

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9. Future Scope

Potential extensions of this analysis include:

- Customer lifetime value prediction
- Personalized recommendation systems
- Real-time dashboard integration with transaction systems